Survey Workbench v2.2

User Manual

October 2026

Version 2.2 – Internal Use Only

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1 Overview

The Survey Workbench is a robust desktop application designed to streamline participant folder generation and automate questionnaire data extraction from fillable PDF forms into centralized CSV or Excel masterfiles.

1.1 Key Features

- Dynamic questionnaire configuration supporting unlimited custom survey types.
- Batch processing capabilities for generating folders and extracting data across multiple participants simultaneously.
- Participant list import functionality supporting text and CSV formats.
- Reusable template bundles for saving and reloading questionnaire setups.
- Automated duplicate detection against masterfiles (CSV, XLS, XLSX).
- Data completeness verification and missing data reporting.
- Interactive help system featuring tooltips and a dedicated "What's This?" mode.
- Automated format detection for masterfiles with interactive preview dialogs prior to committing data.

1.2 System Requirements

- Windows 10/11 or Linux operating systems.
- Microsoft Excel (required for Excel masterfile readout via xlwings on Windows).
- Python 3.x environment with PyQt5 and pypdf dependencies (or deployed as a standalone compiled binary).

2 Getting Started and Interactive Help

When you launch Survey Workbench v2.2, the user interface is cleanly divided into participant generation and data extraction panels. If you are ever unsure what a specific button, text box, or checkbox does:

- **Tooltips**: Hover your mouse over any control to see a brief description of its function.
- **What's This? Mode**: Navigate to **HELP → What's This? (Click for Help)** or press **Shift+F1**. Your cursor will change into a question mark; click on any interface element to open a detailed explanatory window outlining its exact purpose and usage.

3 Step-by-Step Guide: Generating Participant Folders

3.1 1. Declaring Participant IDs

- **Single Mode**: Enter a unique identifier directly into the **Participant ID** text box (e.g., `P001`). This will be used as the subfolder name.
- **Batch Mode**: Check **Batch Mode (multiple IDs)** to open a text area where you can input multiple participant identifiers (one per line or comma-separated). Alternatively, click **Import List...** to load IDs directly from an external text or CSV file.

3.2 2. Setting Target Path and Questionnaire Types

1. Click **Search** next to **Target folder PATH** to choose where participant subfolders will be created.
2. In **Number of questionnaire types**, enter how many different questionnaires each participant needs to complete (e.g., Demographics, NASA-TLX). Click **Confirm** to generate the corresponding configuration rows.
3. For each row, specify the survey name, select the PDF template via the **Search** button, and set the required number of copies.

3.3 3. Using Template Bundles

To save time on recurring studies, you can save your current questionnaire layout by clicking **Save Template Bundle**. Later, you can instantly restore this complete setup across different sessions by clicking **Load Template Bundle**.

3.4 4. Generating and Pre-filling Form Data

Click **Generate Participant Folder**. The software will create the participant subfolders, copy the designated PDF templates with proper naming conventions, and open an interactive prefill preview grid. This allows you to inspect and modify default administrative fields (such as pre-populating participant IDs) before finalizing the files.

4 Step-by-Step Guide: Extracting Data to Masterfiles

4.1 1. Specifying Source and Masterfile Paths

- **Source folder PATH**: Select the experiment folder containing your completed participant subfolders.
- **Masterfile PATH**: Select your target masterfile. Survey Workbench v2.2 automatically detects whether you are using a text-based `.csv` file or an Excel workbook (`.xls` / `.xlsx`).

4.2 2. Quality Control: Completeness and Duplicates

Before running extraction, the software incorporates built-in safety checks:

- **Duplicate Detection**: Automatically scans the masterfile to check if a participant ID has already been extracted, preventing accidental duplicate entries.
- **Data Completeness**: Verifies that all expected readable PDF form fields and data files are present. You can also click **Missing Data Report** to scan the entire source directory and generate a comprehensive summary of complete versus incomplete participant folders.

4.3 3. Field Mapping and Preview

Access **File** → **Save/Edit Field Mapping** from the menu bar to map technical PDF form field identifiers to user-friendly column headers. When extracting data, an interactive preview dialog will display the extracted values, giving you a final chance to inspect and edit them before committing to the masterfile.

5 Configuration Management

You can save your complete application stateincluding folder paths, active questionnaire setups, and field mappingsby going to **File** → **Save Configuration**. Saved configurations can be easily reloaded or deleted using the respective submenus under the **File** menu.

6 Technical Information and Version History

6.1 Complete Version History

- **v2.2 (Current Edition)**: Introduced comprehensive PDF form field extraction via AcroForm and widget annotation scanning, interactive prefill and extraction preview dialogs, robust batch processing with list imports, automated duplicate checking, missing data reporting, and flexible configuration snapshotting.
- **v2.1**: Added structured field mapping capabilities, enhanced masterfile format autodetection, and refined prefill preset handling.
- **v2.0**: Established dynamic questionnaire configuration supporting unlimited questionnaire types, template bundle systems, and unified masterfile readout for CSV and Excel formats.
- **v1.0 (Initial Release)**: Basic static folder generation and single-format CSV data extraction for fixed questionnaire types (Demographics and NASA-TLX).

6.2 Technology Stack and Deployment

- Python 3.x
- PyQt5 (GUI framework featuring tooltips and What's This interactive help)
- xlwings (Excel integration and workbook automation)
- pypdf (PDF form field reading and writing)
- Nuitka (Cross-platform compiler used to compile the source code into native standalone executables, generating '.exe' binaries on Windows and ELF binaries on Linux)